Economic Growth and Carbon Emissions Decoupling

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2026-10-25

Topic Summary

As global economies continue to grow, a fundamental challenge for Sustainable Development Goal 13 (Climate Action) is whether nations can expand their economic output without a corresponding increase in their carbon footprint. This project investigates the historical relationship between national economic performance (GDP per capita) and fossil fuel carbon emissions over the past two decades. By exploring spatial trends and temporal trajectories across different income groups, the analysis aims to determine whether genuine "decoupling" of economic development from carbon emissions is occurring globally. Statistical graphs and trend lines will be used to highlight structural patterns in clean energy adoption and emissions intensity.

Data and Links

Data 1: CO2 Emissions per Capita vs. GDP per Capita Data Link: https://ourworldindata.org/grapher/co2-emissions-vs-gdp

Data 2: Carbon Intensity: CO2 Emissions per Dollar of GDP Data Link: https://ourworldindata.org/grapher/co2-intensity

Data 3: Annual CO2 Emissions by Country Data Link: https://ourworldindata.org/grapher/annual-co2-emissions-per-country

Data 4: GDP Per Capita (World Bank Data via OWID) Data Link: https://ourworldindata.org/grapher/gdp-per-capita-worldbank

Main Data Source

Our World in Data (OWID): https://ourworldindata.org/